My first idea was smart glasses that assist those whose vision are impaired. Ie, they have a camera for OCR so that they can recognise and read text aloud ~~as well as a microphone that can recognise speech and project it onto a small OLED screen on the front.~~

However, upon further consideration this idea *is not unique*. Therefore, introducing a context-aware AI system would probably distinguish it. What I mean by that is, for instance, **if you were in a pharmacy it would read the medicine names and filter out irrelevant details. Likewise, if you were at a bus stop it could focus on only the arrival times, numbers, etc. (important stuff).**

~~In addition, for the hearing-impaired users, we could train the AI to detect tone as well, which would be a bit of a step up from speech-to-text.~~

However, *practically* this is a little hard to execute, I thought. Then, I kinda realised **the only hard bit is the hardware**, once we cobbled together a decent prototype we could easily add other AI features eg. summarise doc, etc...

The challenge is the hardware, software is easy. I think that we can certainly execute this – it isn't very hard.

DESIGN PHILOSOPHY

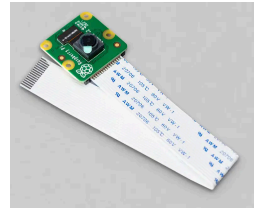
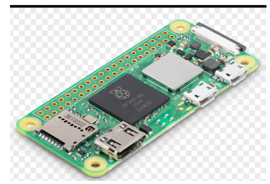
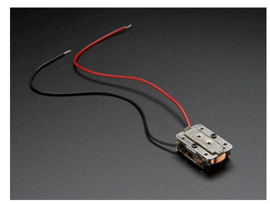
Trying to make the Pi do everything will result in failure and a burnt Pi. Instead, the user can tap the frame, the image will be captured, OCR performed and displayed on screen. We therefore need:

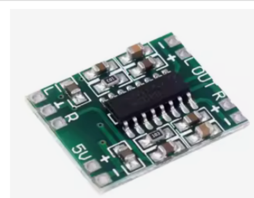
- Trigger-based capture
- Mode-based processing
- Lightweight AI

PARTS

The hardware is the hardest part. Software is always doable.
Also, using BONE CONDUCTION as the speakers – these are little vibration thingies that we place against a skull or bone and they vibrate tricking the brain into *thinking* they are sound whilst they aren't.

BOM

Part	Purpose	Model	Image	Cost (£)
Camera	Capture visual -> OCR	RPi cam v2 - Even though this failed us last time, it is cheap and good - we just need to make sure we are wiring it properly. This camera is good enough for OCR		15
Compute	Control Everything	RPi 02w - controls everything, cheap, quad-core, ok for AI on demand		20
Speaker	Play sound (visually impaired)	Bone Conductor Transducer with Wires - 8 Ohm 1 Watt		9
Display	Shows text up-close.	Small 0.96" i2c OLED mounted prism-based?		1
Power	Powers device	3.7V 2000mAh LiPo - cheap but only 1.5/2h of active use.		2

Part	Purpose	Model	Image	Cost (£)
Boost module	Need to convert to 5V as Pi only takes stable 5V	MT3608		1
Frame	Hold everything together	3D Printed	///	1
Pushbutton	Triggers AI	Generic standard component pushbutton		0.05
Speaker amplifier	ADC (we did this in comps sci cmon)	PAM8403		1
TOTAL	The total is	pretty good at around	only around	£50.05

SOFTWARE

LOGIC FLOW

- Button pressed
- Image captured
- OCR ran
- Text extracted
- Based upon context, relevant information selected
- Displayed on screen
- Spoke out loud from mic.

"CONTEXT" MODE

Basically, we will define three contexts.

- Pharmacy
- School
- uhh Bus stop
  When they are selected we focus on the things, eg
- Pharmacy - medicine names
- School - the board
- Bus Stop - time table, bus numbers etc. - this could be based of what time it is too - this is good as it's easy!
  Each mode applies
- kw filters
- regex
- priority ranking

Add logging which is lightweight too

This
is certainly feasible.